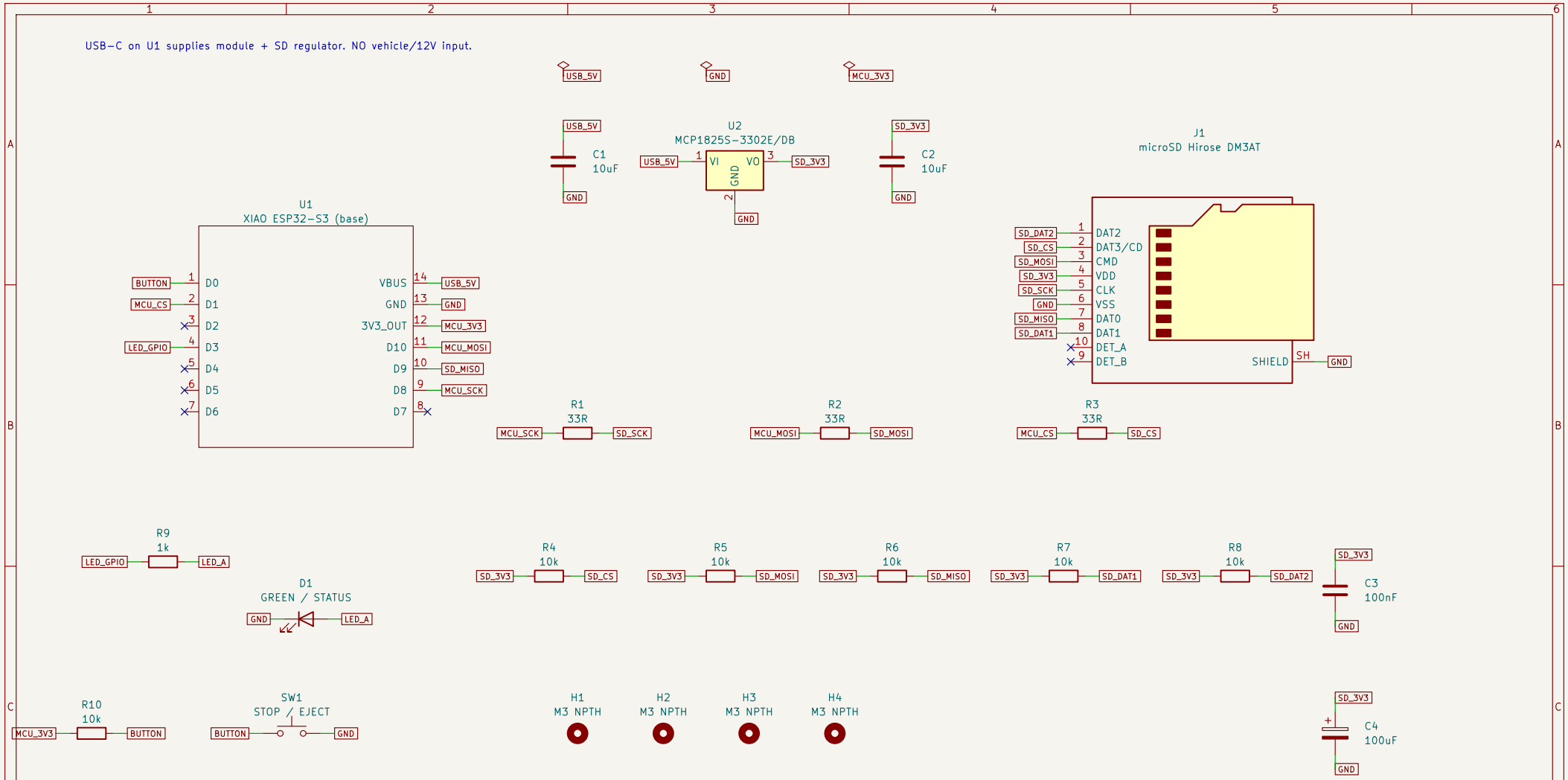

USB-C on U1 supplies module + SD regulator. NO vehicle/12V input.
USB_5V
GND
MCU_3V3
U1
XIAO ESP32-S3 (base)
BUTTON
MCU_CS
LED_GPIO
D0
D1
D2
D3
D4
D5
D6
VBUS
GND
3V3_OUT
D10
D9
D8
D7
USB_5V
GND
MCU_3V3
MCU_MOSI
SD_MISO
MCU_SCK
C1
10uF
U2
MCP1825S-3302E/DB
VI
GND
VO
SD_3V3
C2
10uF
J1
microSD Hirose DM3AT
SD_DAT2
SD_CS
SD_MOSI
SD_3V3
SD_SCK
GND
SD_MISO
SD_DAT1
DAT2
DAT3/CD
CMD
VDD
CLK
VSS
DAT0
DAT1
DET_A
DET_B
SHIELD
SH
R1
33R
MCU_SCK
SD_SCK
R2
33R
MCU_MOSI
SD_MOSI
R3
33R
MCU_CS
SD_CS
R9
1k
LED_GPIO
LED_A
D1
GREEN / STATUS
R4
10k
R5
10k
R6
10k
R7
10k
R8
10k
C3
100nF
R10
10k
SW1
STOP / EJECT
H1
M3 NPTH
H2
M3 NPTH
H3
M3 NPTH
H4
M3 NPTH
C4
100uF
SPI mode: DAT0=MISO, CMD=MOSI, DAT3=CS. DET_A/B intentionally unused.
Separate MCU_3V3 and SD_3V3: do not bridge. All logic3.3V.
Press STOP; wait for solid SAFE LED before removing card or USB.
External antenna must be connected; keep radiating antenna outside PCB/metal enclosure.
Local prototype — no vehicle electrical connection
Sheet: /
File: chrysler_ble_sd.kicad_sch
Title: Chrysler BLE-to-SD USB carrier
Size: A4
Date:
Rev: A
KiCad E.D.A. 10.0.7
Id: 1/1